PORTFOLIO CASE STUDY

Construction Change Order and RFI Analytics

Root-cause, cycle-time and impact analysis for construction decision workflows

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Analytics lifecycle: Ask · Prepare · Process · Analyze · Share · Act

August 2026 · 90 projects · 3,318 RFIs · 1,119 change orders · 2022–2025

Synthetic-data disclosure All project names, organizations, people, budgets, schedules, RFIs, changes, and performance records are fictional. The case study does not represent actual In Project LLC clients or confidential construction records.

Contents

1. Executive Summary

This case study analyzes construction RFI and change-order workflows as connected decision systems. The purpose is to identify where response, review, revision, approval, and forecast processes create commercial and schedule exposure and to convert those findings into an implementation-ready management plan.

Portfolio	RFI Performance	Change Performance	Workflow Health
90 projects 3,318 RFIs 1,119 changes	12.21 avg. days 45.7% on time 295 overdue open	$204.57M approved $88.10M pending 28.91 avg. days	29 Red 48 Yellow 13 Green

Principal findings

- Project-average RFI response and project-average change approval cycle produced the strongest tested portfolio relationship: Pearson r = 0.817 and R^2 = 0.668.
- Slow owner-decision projects averaged 16.45 RFI-response days and 37.98 change-approval days.

- High digital-coordination projects averaged 9.55 RFI-response days and 62.0% on-time performance.
- Coordination Conflict RFIs generated the greatest actual RFI cost impact, approximately $8.23 million.
- Owner-Directed Change, Design Error or Omission, and Unforeseen Condition generated approximately $124.28 million in approved change value.
- Submission-only attributes did not reliably predict later RFI-to-change linkage; the exploratory model is not approved for production use.

2. Business Problem and Analytical Questions

Construction RFIs and change orders move through multiple reviews, handoffs, revisions, pricing steps, and approval gates. When decision workflows are slow or incomplete, projects may experience unpriced work, forecast uncertainty, delayed field decisions, rework, and commercial disputes.

Primary business question

Which change-order causes and RFI workflow conditions create the greatest cost and schedule exposure, and where should management intervene to improve response and approval performance?

Stakeholder decisions

Stakeholder	Decision need
Executive Sponsor	Prioritize intervention and allocate decision resources.
Project Controls Lead	Monitor health, cycle time, exposure, and forecast incorporation.
Design Manager	Improve RFI quality, response accountability, and coordination.
Change Manager	Control pricing, aging, negotiation, approval, and documentation.
Project Manager	Own project recovery actions and unresolved decisions.
Data Analyst	Maintain reproducible metrics, data quality, dashboards, and alerts.

3. Professional Analytics Lifecycle

The project uses the six-phase Ask–Prepare–Process–Analyze–Share–Act structure. Professional terminology from other frameworks is included to show conceptual alignment without treating the models as identical.

Project phase	CRISP-DM	DMAIC	Microsoft	PMI governance
Ask	Business Understanding	Define	Business Understanding	Initiating / Planning
Prepare	Data Understanding	Measure	Data Acquisition and Understanding	Planning

Project phase	CRISP-DM	DMAIC	Microsoft	PMI governance
Process	Data Preparation	Measure / early Analyze	Data Acquisition and Understanding	Executing / Monitoring
Analyze	Modeling and Evaluation	Analyze	Modeling	Executing / Monitoring
Share	Evaluation	Analyze / Improve design	Customer Acceptance	Stakeholder engagement
Act	Deployment	Improve and Control	Deployment / Customer Acceptance	Executing / Monitoring / Closing

The practical workflow is iterative. Data quality, analytical logic, business interpretation, and action planning may require returning to earlier phases.

4. Dataset and Relational Model

A deterministic synthetic dataset was created because public datasets rarely combine project attributes, RFI lifecycle dates, change lifecycle dates, workflow events, impact values, forecast-incorporation dates, and explicit RFI-to-change links.

Table	Clean records	Grain
Projects	90	One row per project
RFI Log	3,318 clean	One row per RFI
Change Orders	1,119 clean	One row per change item
Workflow Events	11,286 clean	One row per transition or handoff
RFI Change Links	677 clean	One row per explicit relationship

Relational structure

Projects relate one-to-many to RFIs, change orders, and workflow events. RFIs and changes relate many-to-many through the RFI Change Links bridge. The bridge prevents direct-join duplication and supports conversion analysis.

5. Data Preparation and Quality

The raw package contained 16,536 records. The Process phase produced 16,490 clean records after removing 28 duplicates and quarantining 18 records whose relationships could not be trusted.

Control	Result

Control	Result
Raw records	16,536
Clean records	16,490
Duplicate removals	28
Quarantined records	18
Cleaning actions	226
Quality checks	22 of 22 passed
Foreign-key violations	0 unresolved
SQLite integrity	ok

Raw files were preserved. Cleaning occurred only in clean and processed layers. Invalid parent relationships were quarantined rather than guessed, and repairs were made only when workflow evidence was unambiguous.

6. Analytical Results

The analysis combined descriptive statistics, aging, cycle-time analysis, Pareto comparisons, workflow-stage duration, project-level transparent risk scoring, correlation tests, and exploratory classification.

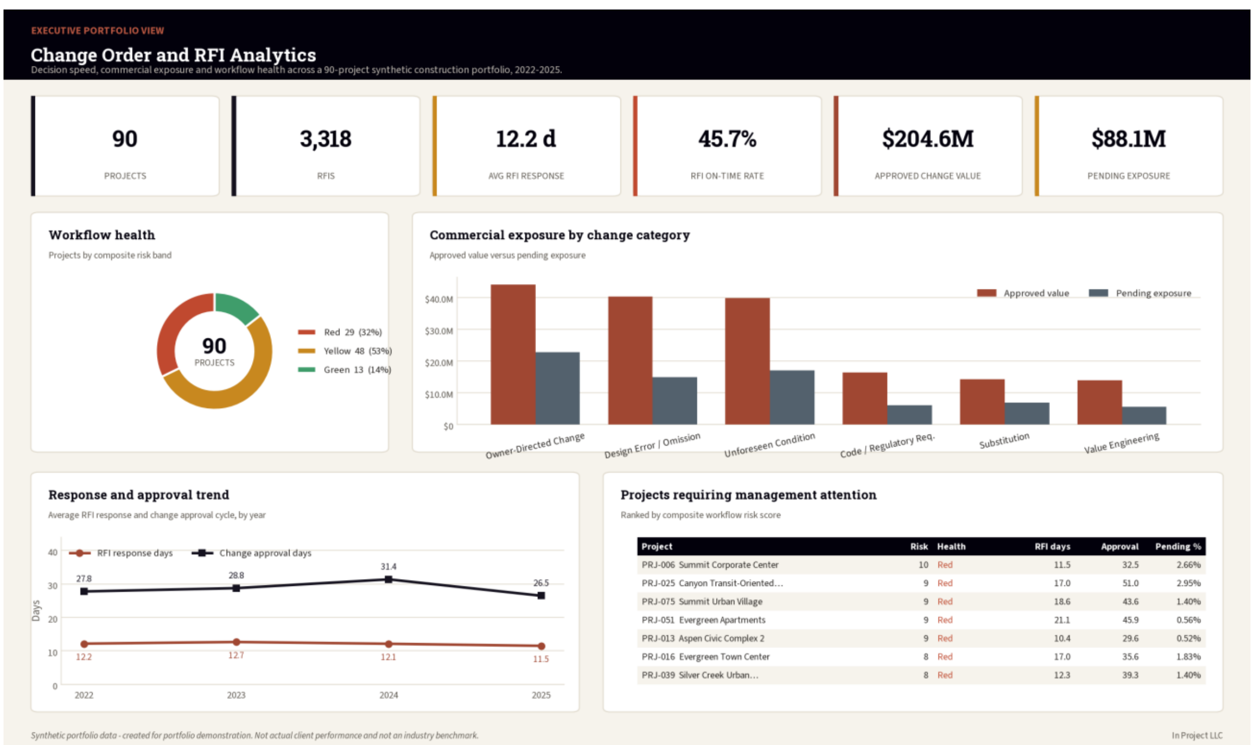


Figure 1. *Executive portfolio view: decision speed, commercial exposure, workflow health and the projects needing attention.*

7. RFI Performance

The portfolio contained 3,318 RFIs. Average final response time was 12.21 days and the on-time rate was 45.7%.

Discipline	RFIs	Avg. days	On-time	Cost impact
Specialty Systems	231	13.64	41.2%	$1.87M
Mechanical	480	13.43	38.7%	$4.94M
Electrical	518	12.98	42.5%	$5.15M
Structural	413	12.95	41.1%	$2.97M
Fire Protection	181	12.41	43.4%	$2.02M
Civil	285	12.38	44.9%	$2.36M

Mechanical, Electrical, Structural, and Specialty Systems displayed longer average response cycles than the portfolio average. The result supports discipline-specific response governance rather than a single undifferentiated backlog target.

8. Change Order and Commercial Exposure

The portfolio contained 1,119 change orders. Approved value was $204.57 million and pending absolute exposure was $88.10 million.

Category	Count	Approved value	Pending exposure	Avg. approval
Owner-Directed Change	198	$44.09M	$22.85M	32.3 days
Design Error or Omission	197	$40.30M	$14.91M	32.9 days
Unforeseen Condition	168	$39.88M	$17.02M	28.4 days
Code or Regulatory Requirement	78	$16.44M	$6.09M	32.0 days
Substitution	94	$14.28M	$6.96M	25.8 days
Value Engineering	93	$13.92M	$5.60M	24.6 days

9. Workflow Bottlenecks

Workflow-event analysis distinguishes volume from duration. A stage may accumulate total time because it is slow, because it has high volume, or both.

Item	Status	Assigned role	Avg. stage days	Total stage days

Item	Status	Assigned role	Avg. stage days	Total stage days
Change Order	Approved	Project Manager	20.31	14335.35
RFI	Closed	Subcontractor	9.93	9411.6
RFI	Closed	General Contractor	9.57	8854.16
Change Order	Returned for Revision	Cost Manager	16.7	8552.62
Change Order	Submitted	Owner Representative	7.03	7850.68
RFI	Returned for Clarification	General Contractor	11.95	4109.42
RFI	Returned for Clarification	Subcontractor	10.86	3345.12
RFI	Answered	General Contractor	9.48	3319.07

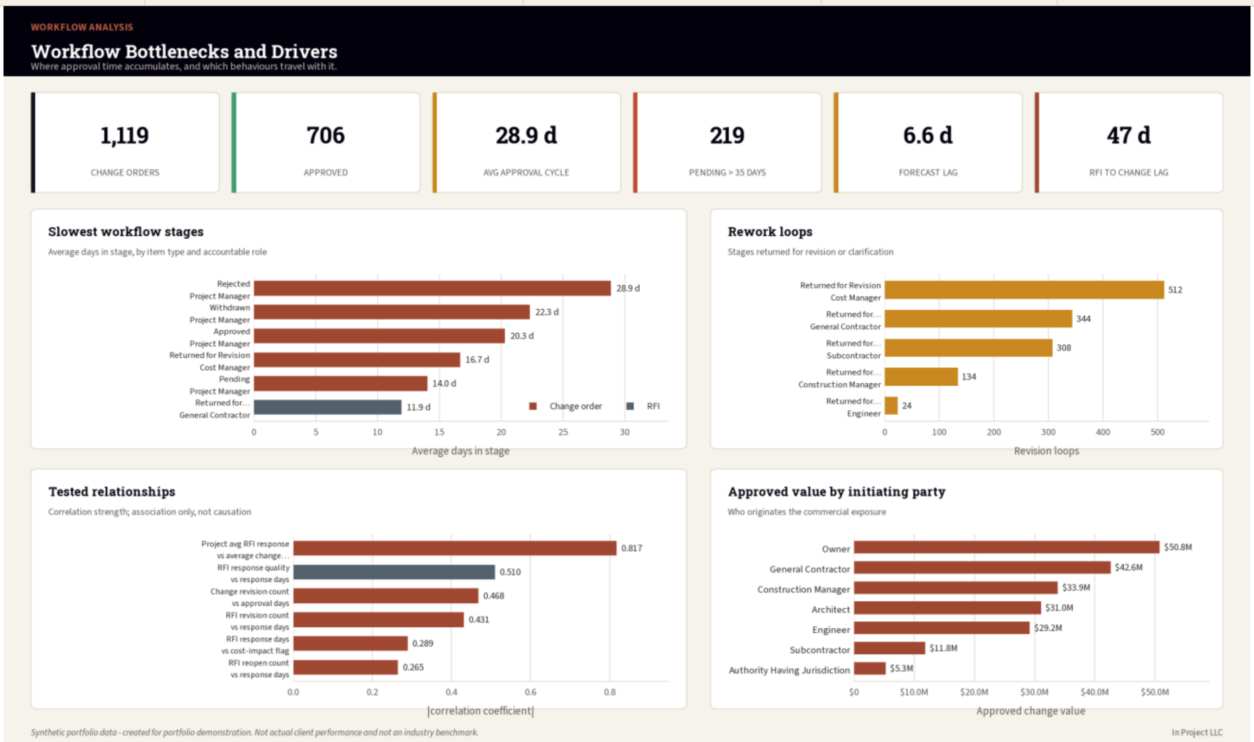


Figure 2. *Workflow bottlenecks and drivers: where approval time accumulates and which behaviours travel with it.*

10. Project Priority and Statistical Evidence

The project-health framework uses seven transparent indicators. Each indicator receives 0 points when Green, 1 when Yellow, and 2 when Red. Overall status is Green for 0–2 points, Yellow for 3–5 points, and Red for 6 or more.

Rank	Project	Score	RFI days	Approval days	Pending / budget
1	Summit Corporate Center	10	11.5	32.5	2.66%
2	Canyon Transit-Oriented Development	9	17.0	51.0	2.95%

Rank	Project	Score	RFI days	Approval days	Pending / budget
3	Summit Urban Village	9	18.6	43.6	1.40%
4	Evergreen Apartments	9	21.1	45.9	0.56%
5	Aspen Civic Complex 2	9	10.4	29.6	0.52%
6	Evergreen Town Center	8	17.0	35.6	1.83%
7	Silver Creek Urban Village 2	8	12.3	39.3	1.40%

Statistical interpretation

Relationship	Method	Statistic	Interpretation
Project avg. RFI response vs avg. approval cycle	Pearson r	0.817	Strong positive portfolio-level association
Project avg. RFI response vs avg. approval cycle	Spearman ρ	0.804	Strong monotonic association
RFI revisions vs response time	Spearman ρ	0.432	Moderate positive association
Change revisions vs approval time	Spearman ρ	0.468	Moderate positive association
Response quality vs response time	Spearman ρ	-0.510	Longer cycles associated with lower quality

Exploratory models

The submission-only logistic model achieved mean cross-validated AUC 0.510, approximately chance level. A diagnostic model using response, revision, quality, and impact indicators improved AUC to 0.631. Neither model is approved for production use.

11. Dashboards and Communication

The Share phase created executive, operational, workflow, and project-priority dashboards, plus Power BI and Tableau build specifications.

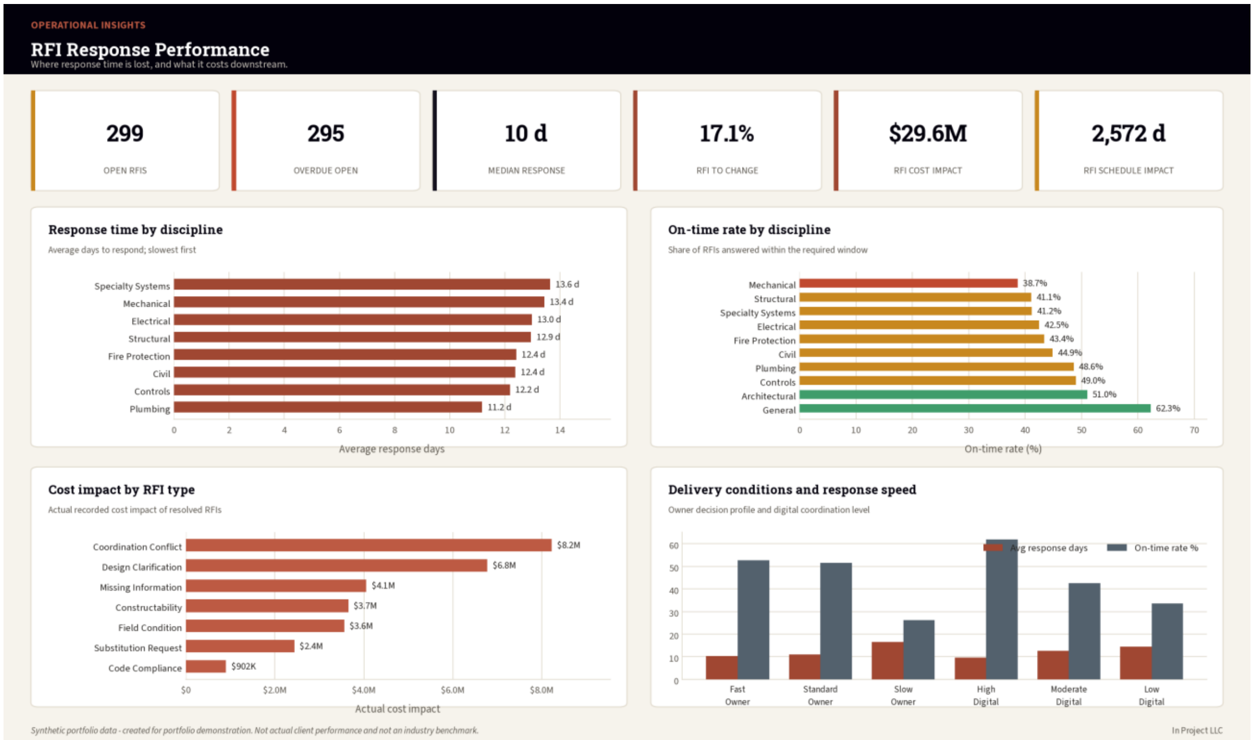
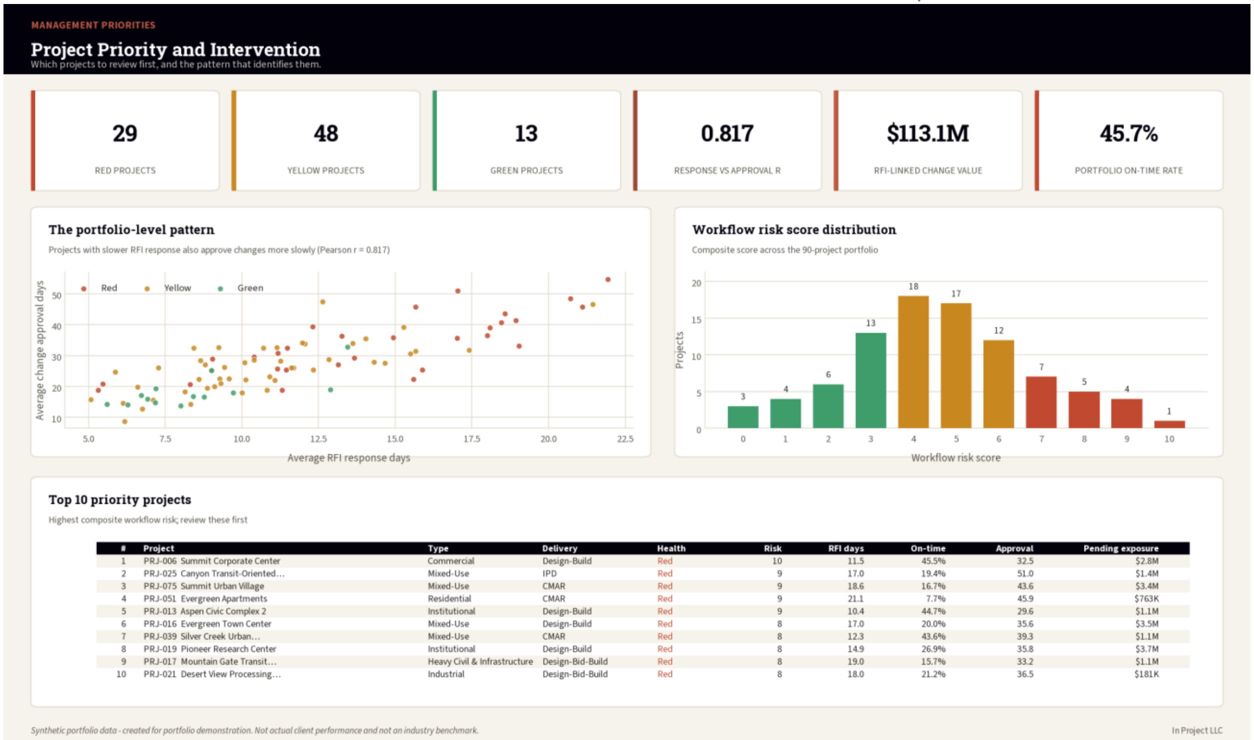


Figure 3. *RFI response performance by discipline and type, with the delivery conditions that shape it.*



#	Project	Type	Delivery	Health	Risk	RFI days	On-time	Approval	Pending exposure
1	PRJ-006 Summit Corporate Center	Commercial	Design-Build	Red	10	11.5	45.5%	32.5	$2.8M
2	PRJ-025 Canyon Transit-Oriented…	Mixed-Use	IPD	Red	9	17.0	19.4%	51.0	$1.4M
3	PRJ-075 Summit Urban Village	Mixed-Use	CMAR	Red	9	18.6	16.7%	43.6	$3.4M
4	PRJ-051 Evergreen Apartments	Residential	CMAR	Red	9	21.1	7.7%	45.9	$763K
5	PRJ-013 Aspen Civic Complex 2	Institutional	Design-Build	Red	9	10.4	44.7%	29.6	$1.1M
6	PRJ-016 Evergreen Town Center	Mixed-Use	Design-Build	Red	8	17.0	20.0%	35.6	$3.5M
7	PRJ-039 Silver Creek Urban…	Mixed-Use	CMAR	Red	8	12.3	43.6%	39.3	$1.1M
8	PRJ-019 Pioneer Research Center	Institutional	Design-Build	Red	8	14.9	26.9%	35.8	$3.7M
9	PRJ-017 Mountain Gate Transit…	Heavy Civil & Infrastructure	Design-Bid-Build	Red	8	19.0	15.7%	33.2	$1.1M
10	PRJ-021 Desert View Processing…	Industrial	Design-Bid-Build	Red	8	18.0	21.2%	36.5	$181K

Figure 4. *Project priority and intervention: the portfolio-level response-versus-approval pattern and the top 10 review list.*

The dashboard design emphasizes decisions required, responsible roles, exposure, cycle time, and evidence. Health is communicated through both color and status text.

12. Act Plan and Management Recommendations

The Act phase proposes a 30/60/90-day operating plan. Targets are management objectives for this synthetic case study and require validation before real use.

Measure	Baseline	90-day target
RFI on-time rate	45.7%	70.0%
Average RFI response	12.21 days	10.0 days
Average change approval	28.91 days	24.0 days
Pending exposure	$88.10M	≤$66.08M
Overdue open RFIs	295	≤148
Forecast incorporation lag	6.58 days	≤5.0 days
Old pending changes >35 days	219	≤131
Red projects	29	≤18

Recommended sequence

- Days 0–30: stabilize exceptions, validate backlog, establish decision forums, and reconcile forecast gaps.
- Days 31–60: reduce backlog, introduce quality gates, review root causes, and pilot digital coordination.
- Days 61–90: standardize reporting, generate governed alerts, close priority actions, and measure benefits.
- Days 91–180: sustain controls, expand successful pilots, and review thresholds with evidence.

13. Governance, Controls, and Automation

The governance model uses daily exceptions, twice-weekly RFI coordination, weekly change and project-health reviews, weekly forecast reconciliation, monthly executive and root-cause reviews, and quarterly methodology review.

Automation output	Count	Severity
Field-Impact Overdue RFI	71	Critical
High/Critical Overdue RFI	84	Critical
Red Project	29	Critical
High Pending Exposure	15	Critical
Approved Not Incorporated	56	High
Forecast Incorporation Lag	299	High
Old Pending Change	219	High

The tested alert script generated 773 synthetic management alerts: 199 Critical and 574 High. This demonstrates exception generation; it does not create a live scheduled service or send external notifications.

14. Limitations and Responsible AI

- All data is synthetic and is not an industry benchmark.
- Thresholds and targets are management assumptions for the case study.
- Associations do not establish causality.
- Project-level aggregation may conceal item-level variation.
- The backlog is intentionally severe in portions of the synthetic dataset.
- The exploratory models require real governed data, temporal validation, leakage controls, calibration, subgroup review, monitoring, and stakeholder acceptance.
- No automated alert should be sent externally until recipients, suppression rules, escalation, and data governance are approved.

15. Conclusion and Sources

The case study demonstrates how construction project-management knowledge and data analytics can be combined to move from raw workflow records to reproducible analysis, executive communication, and implementation-ready controls.

Professional capability demonstrated

- Construction project controls and workflow governance
- Relational data design, cleaning, and validation
- Excel, SQL, SQLite, and Python analytics
- Power BI and Tableau dashboard planning
- Statistical interpretation and model governance
- Management action planning, RACI, KPIs, and benefits realization
- Responsible AI leadership and human oversight

Methodology references

IBM CRISP-DM overview: https://www.ibm.com/docs/en/spss-modeler/18.6.0?topic=dm-crisp-help-overview

ASQ DMAIC: https://asq.org/quality-resources/dmaic

Microsoft Data Science Lifecycle: https://learn.microsoft.com/en-us/shows/dev-intro-to-data-science/what-is-the-data-science-lifecycle-2-of-28

PMI Process Groups: https://www.pmi.org/standards/process-groups

End of Report